

Reusing Cardboard for Packaging Boxes with a Computational Design System

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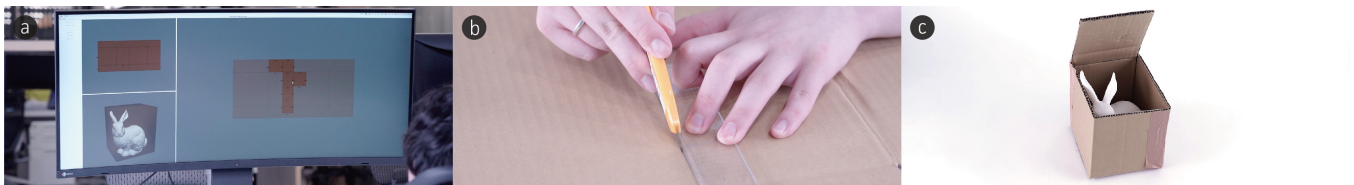


Figure 1: Design and fabrication workflow. (a) The system takes a 3D model of an item to be shipped and the pattern of cardboard to be reused as input, and visually suggests the optimal net for a new packaging box. By cutting and folding the cardboard following the suggestion (b), users can reconstruct an existing cardboard box into a precisely fitting packaging box (c).

ABSTRACT

We propose a computational design system for packaging boxes, which enables reconstruction of cardboard into packaging structures that precisely fit the size of the objects to be shipped. Given a 3D model of the object and the structure of the cardboard to be reused, our system suggests the optimal net of a new box, which only requires minimum cuts and folds for fabrication by leveraging the edges of the input cardboard. To achieve this, we implemented a GUI visualizing the input models and the optimal net computed by our optimization algorithm. As a demonstration, we showed three design examples, with evaluation results of our method for reducing the length of cuts and folds.

CCS CONCEPTS

• **Applied computing** → **Computer-aided design**; *Computer-aided manufacturing*; • **Computing methodologies** → *Optimization algorithms*.

KEYWORDS

personal fabrication, daily object reuse, optimization algorithm, sustainable fabrication, sharing economy

ACM Reference Format:

Kazuki Koyama, Koya Narumi, Ken Takaki, Yasushi Kawase, Ari Hautasaari, and Yoshihiro Kawahara. 2023. Reusing Cardboard for Packaging Boxes with a Computational Design System. In *The 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23 Adjunct)*, October 29–November 01, 2023, San Francisco, CA, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3586182.3616692>

1 INTRODUCTION

With a recent increase in individual-to-individual transactions [5], such as with C2C trading, it is increasingly important for general users as well as professional carriers to find and use packaging boxes with appropriate sizes, both for sustainable delivery and secure transportation. However, our preliminary online survey in Japan ($n = 100$) found that (1) 96% of respondents sometimes or often have trouble finding cardboard of suitable sizes for the shipped objects; (2) more than 85% do not have foam or air cushion at home, which can tightly fill a packaging box for secure shipping; (3) on the other hand, 88% have leftover cardboard at home; (4) 86% can get cardboard for free. Please refer to the poster for more details.

Considering the survey result, we are motivated to build a software system that enables users to make their own packaging boxes and other components like cushions tightly fitting given objects by only using easily available cardboard materials and simple fabrication methods like cutting and folding. As a first step towards this, we propose a GUI that generates a net schematic of the packaging box from a 3D model of an object to be shipped and any cardboard box larger than the object. The key to our system is the optimization algorithm that can minimize the length of cuts and folds to fabricate the box. We demonstrate three design examples and evaluate our algorithm for reducing the length of cuts and folds for fabrication.

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UIST '23 Adjunct, October 29–November 01, 2023, San Francisco, CA, USA

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ACM ISBN 979-8-4007-0096-5/23/10.

<https://doi.org/10.1145/3586182.3616692>

2 RELATED WORK

In HCI, numerous design and fabrication methods have recently been proposed based on a folding process. Such examples include Fibercuit [7] that combines origami transformation and circuitry, Flaticulation [3] that creates 3D shapes with hinges that fold at specific angles, HingeCore [1] that creates 3D shapes by folding laser-cut sandwich structures, and Kyub [2] that enables to design 3D shapes based on box shapes and corresponding laser-cut patterns. Regarding the design software of origami, Crane [6], which performs origami simulations and computation, allows general users to perform dynamic simulation and form-finding of origami patterns.

While our system shares the idea of folding fabrication with such prior works, our system specifically deals with the 2D transformation problem of unfolded shapes. This transformation in the 2D domain constitutes an optimization problem, which cannot be solved due to the non-differentiability of the objective. By reformulating the problem into a mixed-integer programming framework, our system can effectively solve this optimization problem.

3 WORKFLOW

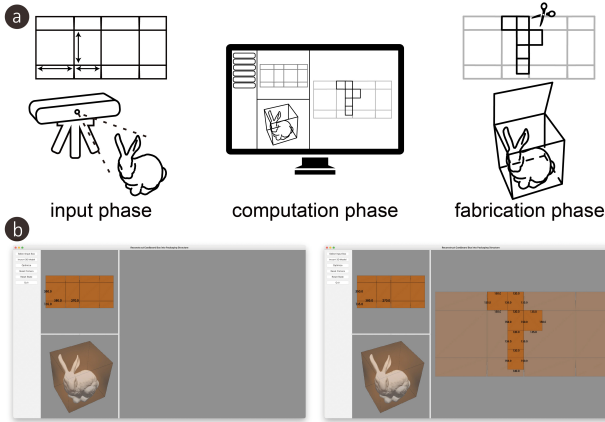


Figure 2: (a) Our system consists of the input phase, computation phase, and fabrication phase. (b) Our software automatically computes and suggests the optimal net and layout that can be folded into a packaging box.

As shown in Figure 2a, users start with the input phase in which they specify the pattern and the size of an original 2D cardboard and the 3D model of an item to be shipped. In the computation phase, our software presents the optimal net and layout among the possible nets that can be folded into the target shape (Figure 2b), by enumerating them by flattening algorithm [4]. In this optimization, in order to minimize users' effort, our system suggests a net that requires minimum cuts and folds for fabrication, resulting in the maximization of the common parts of the boundary and inner edges between the input net and the target net. By solving this maximization problem, our system suggests to users the theoretically optimal net and layout. In the fabrication phase, users can manufacture the packaging box by cutting the layout from the cardboard and assembling it according to the suggestion.

4 RESULT

Fabrication result. We demonstrated our system with the design examples shown in Figure 3. The optimal layouts vary for each original box and object, ensuring each object is packed in a new box that fits the size of the object.

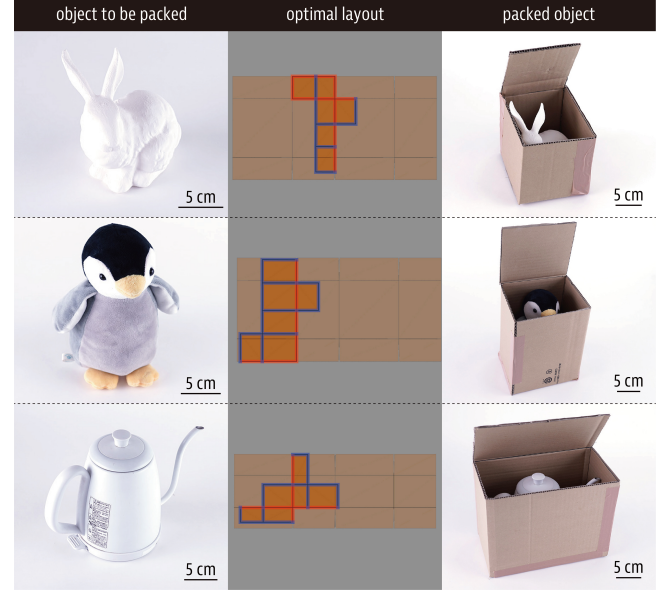


Figure 3: Examples of reconstruction with our system. Each column shows the object to be packed, the optimal layout of the target net, and the packed object. In the column of layout, common edges between the input and the target net are highlighted in red, and the cuts and folds are in blue. Each row shows the example of a 3D printed bunny, a penguin plush toy, and a kettle.

Optimization result. To evaluate our optimization process, we compared our optimization algorithm with the baseline algorithm based on the 'place in the top left' approach. As a comparison metric, we compared the ratio of the total length of lines to be newly cut or folded (i.e., blue lines in Figure 3) against the length of all edges of the target net. As a result, the ratio was successfully reduced from 84% to 47% for the bunny. Similarly, it decreased from 84% to 70% for the penguin plush toy and from 84% to 62% for the kettle.

5 CONCLUSION AND FUTURE WORK

We proposed a reconstruction system of cardboard into packaging boxes and demonstrated our workflow with implemented software and design examples. We are planning to perform a user study to investigate our system's usability and sustainability contribution. We will also diversify target shapes, such as boxes including components with cushioning properties, and improve optimality with additional panels or interactive editing of users.

ACKNOWLEDGMENTS

This research is part of the results of Value Exchange Engineering, a joint research project between R4D, Mercari, Inc. and the RIISE.

REFERENCES

- [1] Muhammad Abdullah, Romeo Sommerfeld, Bjarne Sievers, Leonard Geier, Jonas Noack, Marcus Ding, Christoph Thieme, Laurenz Seidel, Lukas Fritzsche, Erik Langenhan, Oliver Adameck, Moritz Dzingel, Thomas Kern, Martin Taraz, Conrad Lempert, Shohei Katakura, Hany Mohsen Elhassany, Thijs Roumen, and Patrick Baudisch. 2022. HingeCore: Laser-Cut Foamcore for Fast Assembly. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology* (Bend, OR, USA) (UIST '22). Association for Computing Machinery, New York, NY, USA, Article 10, 13 pages. <https://doi.org/10.1145/3526113.3545618>
- [2] Patrick Baudisch, Arthur Silber, Yannis Kommuna, Milan Gruner, Ludwig Wall, Kevin Reuss, Lukas Heilman, Robert Kovacs, Daniel Rechlit, and Thijs Roumen. 2019. Kyub: A 3D Editor for Modeling Sturdy Laser-Cut Objects. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland Uk) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–12. <https://doi.org/10.1145/3290605.3300796>
- [3] Chiao Fang, Vivian Hsinyueh Chan, and Lung-Pan Cheng. 2022. Flaticulation: Laser Cutting Joints with Articulated Angles. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology* (Bend, OR, USA) (UIST '22). Association for Computing Machinery, New York, NY, USA, Article 7, 16 pages. <https://doi.org/10.1145/3526113.3545695>
- [4] W. Liu and K. Tai. 2007. Optimal design of flat patterns for 3D folded structures by unfolding with topological validation. *Computer-Aided Design* 39, 10 (2007), 898–913. <https://doi.org/10.1016/j.cad.2007.05.015>
- [5] Ministry of Land, Infrastructure and Transport of Japan. 2021. *Transition in the number of parcels handled in the past 36 years (written in Japanese)*. Retrieved July 18, 2023 from <https://www.mlit.go.jp/report/press/content/001494501.pdf>
- [6] Kai Suto, Yuta Noma, Kotaro Tanimichi, Koya Narumi, and Tomohiro Tachi. 2022. Crane: An Integrated Computational Design Platform for Functional, Foldable, and Fabricable Origami Products. *ACM Trans. Comput.-Hum. Interact.* (dec 2022). <https://doi.org/10.1145/3576856> Just Accepted.
- [7] Zeyu Yan, Anup Sathya, Sahra Yusuf, Jyh-Ming Lien, and Huaishu Peng. 2022. Fibercuit: Prototyping High-Resolution Flexible and Kirigami Circuits with a Fiber Laser Engraver. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology* (Bend, OR, USA) (UIST '22). Association for Computing Machinery, New York, NY, USA, Article 12, 13 pages. <https://doi.org/10.1145/3526113.3545652>